

Andrew Klesh

andrew.t.klesh@jpl.nasa.gov

RESEARCH INTERESTS

- In-situ exploration of extreme environments (deep space, arctic, underwater)
- Low-cost and NanoSpacecraft mission development and operations
- Robotic assets & instruments to advance human and space science exploration objectives
- Optimal path-planning and estimation, orbital mechanics, mission system optimization

EDUCATION

Ph.D. Aerospace Engineering (Flight Dynamics and Control)

University of Michigan, Ann Arbor, MI (2007-2009)
Dissertation: "Optimal Exploration Systems"

M.S. Geology

California State Polytechnic University, Pomona, CA (2021-2023) *Thesis: "Improving Knowledge of Lava Pathways at Kilauea Using Distributed Acoustic Sensing"*

M.Eng Space Systems Engineering

University of Michigan, Ann Arbor, MI (2008-2009)

M.S.E. Aerospace Engineering (Flight Dynamics and Control)

University of Michigan, Ann Arbor, MI (2005-2006)

B.S.E. Aerospace Engineering

University of Michigan, Ann Arbor, MI (2001-2005)

B.S.E. Electrical Engineering

University of Michigan, Ann Arbor, MI (2001-2005)

PROFESSIONAL EXPERIENCE

Jet Propulsion Laboratory (2011-Present)

- **Associate Director / Chief Systems Engineer - Caltech/JPL Brinson Exploration Hub (2024-Present)** - Support implementation teams in execution of quick-turn project concepts, from formulation through deployment or launch. The Hub serves as a "skunkworks"-like endeavor for breakthrough science or societal-benefit projects serving the Caltech / JPL community strategic interest and partnering with novel collaborators.
- **Project Systems Engineer – Lunar Trailblazer (2020-2025)** – Technical lead and NASA Engineering Technical Authority for ~200kg lunar orbiting spacecraft. Mission was to map water ice on moon using push-broom spectrometer and thermal mapper. Coordinated multiple institutions for spacecraft integration and mission readiness including international partners. Cost constrained within NASA's SIMPLEX program. Launched Feb 2025.
- **Chief Engineer – MarCO (Mars Cube One) Nanospacecraft (2014-2020)** – Technical lead for dual 6U CubeSat mission to Mars. Spacecraft were the first nanospacecraft to independently leave Earth's orbit and fly by Mars. CubeSats launched with InSight, separated, cruised (including 5 trajectory correction maneuvers), and reached Mars in Nov 2018. Craft received InSight EDL UHF data, and re-transmitted to the DSN's 70m dish on Earth. Project was cost constrained, and had limited schedule (concept to flight in 15 months). Additionally, Cognizant Engineer or Contract Technical Manager of several subsystems (Electrical, CDH, EPS, GDS, and Cameras).
- **Engineering Lead – Buoyant Rover for Under-Ice Exploration (2011-Present)** – Co-designed and constructed novel buoyant under-ice rover for the traversal of arctic and Antarctic extreme environments. Primary operational use has been the investigation of methane seeps in permafrost lakes near Barrow, AK (Over 20 field campaigns completed, additional planned). Able to anchor in place for multiple days, provides video and instrument data. Rover is now remotely controllable via satellite from anywhere in the world for underwater operations. Also supporting long duration under-ice

video/environmental sensor systems, and robotic kite-cams for aerial views.

- **JPL Engineering Lead - Ocean Exploration Vehicles (2016-Present)** – Technical lead of novel systems to explore deep ocean and trenches in partnership with Woods Hole Oceanographic Institution. Includes Orpheus-class vehicles, Mini-Sub, other distributed architectures, and optical & sampling instrumentation.
- **Principal Investigator – INSPIRE: Interplanetary NanoSpacecraft Pathfinder In a Relevant Environment (2012-2014)** – Led dual CubeSat mission to bring NanoSpacecraft to interplanetary space and demonstrate survivability, communications, navigation, COTS component monitoring, and novel science payloads. Team also developed micro-controller-based fault-tolerant software system (in conjunction with standard NASA MOS/GDS). Mission was selected for launch via NASA CLI - ready for flight, June 2014. Additional roles include Chief Engineer and MOS/GDS Lead. Funded by JPL and NASA Science Mission Directorate.
- **Mission Architect / System Engineer / Proposal Manager (2011-Present)** – Support, provide analysis, and develop Earth and planetary mission concepts. Have led or supported proposals for ISS payloads, CubeSats, Small Spacecraft, Small-Body explorers, Interplanetary NanoSpacecraft, Jovian explorers / landers, Discovery & New Frontier missions, robotic-support for manned missions, etc. Provided analysis in system engineering, component selection, radiation analysis, orbital mechanics, and communication systems. Supported Europa Lander activities for sample handling mechanism and extremely low-risk planetary protection incendiary device.
- **CubeSat / NanoSpacecraft Consultant** – Support design, construction and operations of NanoSpacecraft and CubeSats including M-Cubed/COVE (1&2), GRIFEX, CHARM/RACE, INSPIRE, RAX (1&2), ISARA, MarCO, NEAScout, Lunar Flashlight, ASTERIA, RainCube, AOSat, LunaH-Map, Sunrise

ACADEMIC APPOINTMENTS

California Institute of Technology (2017-Present)

Lecturer, Department of Electrical Engineering

- Created and taught laboratory class (with project and seminars) Electronics for Space Applications for undergraduate and graduate students. Two term course providing background in the design and integration of basic electronic payloads for low-cost space systems. Students characterize environment and test payloads through two high-altitude balloon flights. “Project might be one of the coolest things you do at Caltech” – Student survey, Spring 2019.

Lecturer, Department of Aerospace (2017-2018)

- Co-taught class on space systems engineering for undergraduate and graduate students. 1st quarter lectures followed by project 2nd and 3rd quarters.

Visiting Associate, Division of Geological and Planetary Science

Arizona State University (2014-Present)

Adjunct Professor, School of Earth and Space Exploration

- Collaborate on research and proposals for solar system exploration. Created and taught 2015-2022 Field (MacGyver) Engineering Workshop. Consultant on AOSat and LunaH-Map CubeSat missions (including Review Board Chair through CDR). Numerous proposal development activities, invited presentations and reviews, etc.

PREVIOUS RESEARCH HISTORY

Postdoctoral Research

JAXA Space Exploration Center (JSPEC) (2010-2011)

- Analyzed trajectory of DCAM imagers to find Ikaros solar sail warping and distortion. Robotic multi-agent architecture research for deep space exploration. Member: Hayabusa, Ikaros mission teams. Additional work on Akatsuki.

RAX CubeSat (Co-Manager) (2009-2010+)

- Management of ADCS, EPS, and Position & Timing Systems. Design, Hardware, Software, Testing and Operations. Built web-based ground-station and operations tools. Supported high-altitude balloon flights.

Graduate Research (University of Michigan)

- Autonomous information-collection systems and optimal path planning
- Design of perpetually-flying solar-powered aircraft for Mars
- Optimal attitudes for asteroid fly-bys
- Tactical path planning for submarine explorers equipped with sonar
- Information collection with multiple-rover system with sonar sensors
- Range-based information collection for aerial systems
- NASA/JPL planetary science summer school – Trojan Tour (2008)
- NASA Microgravity Flights – Prevention of air embolism in microgravity (2004-2006)

Undergraduate Research

- Micro-UAV Design and Construction (2003-2006)
- NASA Get-Away Special (STS-106) – Space exposure experiments (1998-2001)

ADDITIONAL INDUSTRIAL EXPERIENCE

- **Jet Propulsion Laboratory** – Visiting Researcher (Summer 2008): Investigated optimal path planning algorithms for underwater sonar vehicles.
- **Raytheon Missile Systems, Tucson, AZ (Summer 2006)** – Post-Graduate Intern: Developed tools to autonomously create state-space models.
- **bioMerieux, Inc, St. Louis, MO (Summers 2001-2003)** – Systems Engineering Intern: Manufacturing biomedical devices. Fixes in active assembly line environment.
- **Sakai Foundation, Ann Arbor, MI (2005-2008)** – Deployed and managed network/presentation technology for international conferences.
- **The Millennium Project, Ann Arbor, MI (2002-2008)** – Developed novel educational technology, including 3D tour of historical Michigan campus.

PROFESSIONAL DISTINCTIONS (Individual)

2023JPL Voyager Award – Underwater Tech Development & Indigenous Peoples Involvement 2023JPL Voyager Award – Lunar Trailblazer TVAC Completion 2022JPL Voyager Award – Lunar Trailblazer Leadership 2021JPL Voyager Award – Lunar Trailblazer Project Systems Engineering 2020JPL Voyager Award – Lunar Trailblazer Spacecraft Transition 2020AIAA Engineer of the Year 2019NASA Exceptional Engineering Achievement Medal 2018JPL Voyager Award – MarCO Development and Operations 2018JPL Researcher on Campus (JROC) 2018JPL Voyager Award – MarCO Launch and Tech Demo Completion 2016NASA Astronaut Candidate Finalist 2016JPL Lew Allen Award for

Excellence 2016JPL Voyager Award – MarCO Integration and Test 2015JPL Bonus Award – MarCO Implementation 2014JPL Distinctive Talent 2014JPL Mariner Award – INSPIRE Leadership 2013NASA Astronaut Candidate Finalist 2013JPL Ranger Award – Interplanetary NanoSpacecraft Leadership 2010-11JSPS Postdoctoral Fellowship – Japanese Space Agency (JAXA) 2010IAF Luis G. Napolitano Award 2010NASA Ambassador – Cohort 2 2009NASA Scholar – International Astronautical Congress 2009AAS Breakwell Award 2006-09DoD National Defense Science and Engineering Graduate Fellowship 2008-09AIAA Foundation Guidance Navigation and Control Graduate Award 2008JPL/Caltech Student-Faculty Program (NASA SpaceGrant) 2008NASA/JPL Planetary Science Summer School 2005-06University of Michigan Rotorcraft Design Center Fellowship 2006James B. Angell Scholar 2005Aerospace B.S.E cum laude 2005EECS B.S.E. cum laude 2004-05University Honors 2004-05Dean’s List 2000Vigil Honor 1997Eagle Scout

PROFESSIONAL DISTINCTIONS (Group/Team)

2020Space Foundation Jack Swigert Award for Space Exploration – MarCO 2019Aviation Week & Space Technology – Space Platform Laureate - MarCO 2019Creative Arts Emmy - Outstanding Original Interactive Program - Contributor 2019AIAA SmallSat Mission of the Year - MarCO 2019NASA Group Achievement Award – InSight EDL Communications 2019NASA Group Achievement Award – MarCO Project Team 2019Collier Trophy Finalist – MarCO Mission 2017JPL Team Award - Europa Lander Technology/Concept Maturation 2016JPL Research Poster Award – Buoyant Rover 2015NASA Group Achievement Award – NEAScout Mission Concept Review 2014NASA Group Achievement Award – Buoyant Rover Team 2013NASA Group Achievement Award – MSL Launch/Cruise/Approach Team 2012NASA Group Achievement Award – M-Cubed/COVE CubeSat Flight Team 2012JPL Team Award – Dusty Plasma Physics Proposal Leadership 2012JPL Team Award – CubeSat Development 2009IAA Planetary Defense Student Competition – 3rd Place

TEACHING EXPERIENCE

California Institute of Technology – Instructor

- Deep Space Operations for Small Spacecraft – Approx. 300 students (2021)
- Student Space Operations Center – Approximately 8 students (2019)
- Practical Electronics for Space Applications – 8, 16, 24, 21 students (2017, 2019, 2022, 2026)
- Space Systems – Approximately 25 students (2017, 2018)

Arizona State University – Instructor / Creator

- Field Engineering Workshop (aka MacGyver Engineering) (2015-2022)

University of Michigan – Graduate Student Instructor

- Avionics and Aerospace Software Engineering (Winter 2007)
- Introduction to Experimentation and Instrumentation (2005-2006)
- Introduction to Electronics & Microcontrollers (2006-2009)

Instructor – American Red Cross (2007-2010): Wilderness First Aid / CPR / First Aid for the Professional Rescuer

MENTORING HISTORY

2023Edward Vanderfeen (Australian National Indigenous Space Academy Mentor) 2022Kyle Zahney (JPL MUREP/MSI Program) 2020Noah Gans (JPL Innovation to Flight Internship) 2018Jake Kosberg (Academic Part Time) 2018MarCO Interns – 2 students 2017Dr. Anne Marinan (Early Career Hire) 2016-17NASA Space Technology Research Fellow Collaborator 2014-16Dr. Simon Tardivel (NASA Postdoctoral

Program) 2016Buoyant Rover Internships – 2 students 2016Europa Lander Analogue Ice Manufacturing – 1 student 2015MarCO Internship – 1 student 2013JPL CubeSat Groundstation Development – 1 student 2013INSPIRE Internships – 4 students 2013NanoSpacecraft Landers on Small Body Objects – 1 student 2012GRIFEX Payload Development – 1 student 2012Tracking of LEO CubeSats using Amateur Groundstations – 1 student 2011Development of CubeSat Component Database – 11 students

CERTIFICATIONS & TRAINING

- **NOLS:** Alaska Mountaineering 2016, Mt. Denali Expedition 2019
- **Private Pilot:** IFR rated (Additionally tailwheel and ski-plane trained)
- **Emergency Maneuver Training:** (Unusual Attitudes – Spins, Stalls and Recovery)
- **Scuba:** PADI/NAUI/SDI Rescue (+ Drysuit, Nitrox, Ice-Diving, Altitude)
- **California Science Center Aquarium Dive Team (2013-2017)**
- **Ham Radio:** Technician
- **Medical:** EMT (NREMT / LA County) + IV/Blood Draw
- **WEMT:** Wilderness Emergency Medical Technician (NOLS)
- **DFA-Pro:** Diver's Alert Network Diver First Aid for the Professional Diver
- **USC Hyperbaric Chamber Operator / Supervisor:** Catalina Island
- **USGS Hawaiian Volcano Observatory Volunteer (2017-2020)**

ADDITIONAL ACTIVITIES

- Chicago Marathon (2003, 2004)
- Human Powered Submarine Team (2001-2005)
- Urban Search and Rescue Team – JPL (2008, 2011-Present, Co-Captain)
- Undersea Voyager Project (2009-2010)
- XSAS NASA RGSOP Microgravity Experiment (2010) – Co-PI
- SolarBubbles – Solar-Powered UAV Team (Founder, Team Lead, Advisor) (2005-2010)
- Fuel Cell UAV – World Record Flight (Advisor) (2008)
- Flying Fish – Persistent surveillance sea-plane UAV – 2007-2008
- AIAA GN&C Technical Committee Member – 2008-2011
- Challenger Learning Center Space Camp Teacher (2014-2017)
- Lake Tahoe Ironman (2015)
- Dawn-to-Dusk 12-hour Adventure Race (2017)
- Lake Tahoe MedWars Adventure Race (2018)
- NASA JSC Microgravity NExT Project Lead (2017-2019)
- Santa Cruz Half-Ironman (2025)
- Pasadena Fire Department EMS Reserve (4400+ hours) (2018-Present)
- Catalina Hyperbaric Chamber Volunteer Staff (400+ days) (2019-Present)

INVITED TALKS & PANELS

Oct 2023AIAA ASCEND Panel on Future of Deep Space Small Spacecraft Mar 2018IEEE Aerospace Panel on Science for Small Spacecraft Nov 2016National Academies Arab-American Frontiers Forum Jan 2015AstroRecon Panelist: New Directions in Asteroid Exploration Oct 2020ASU Innovation in Engineering Panel Oct 2020NEPRIS Space Week Oct 2020[Keynote] Michigan Space Grant Consortium Oct 2020MIT Small-Sat Center Monthly Sep 2020CubeSat Launch Workshop - U. Southern Maine Jul 2020NEPRIS Online Student Camp Jun 2020Lincoln Labs - Boston, MA Jun 2020JPL Summer Intern Program Mar 2020Goddard Spaceflight Center Oct 2019[Keynote] IAC Symposium on Small Spacecraft Sept 2019SESE Colloquium,

- R (2024). "Four Thousand Meters Under Ice: Exploring Ice-Covered Chemosynthetic Ecosystems at the Aurora Vent Field with the NUI Hybrid Remotely Operated Vehicle." *2024 Astrobiology Science Conference* [[link]](<https://agu.confex.com/agu/abscicon24/meetingapp.cgi/Paper/1499213>)
- **Klesh, A.** and Polet, Jascha and Zhan, Zhongwen (2023). "Improving Knowledge of Lava Pathways at Kilauea Using Distributed Acoustic Sensing." *American Geophysical Union* [[link]](<https://agu.confex.com/agu/fm23>)
 - Hardgrove, Craig J and Starr, Richard D and Prettyman, Thomas H and Lazbin, Igor and Johnson, Erik Bjorn and Roebuck, Bob and DuBois, Joe and Struebel, Nathaniel and Colaprete, Anthony and Hailey, Patrick and **Klesh, A.T.** and others (2023). "The Lunar Polar Hydrogen Mapper (LunaH-Map) Mission, Launch and Early Operations." *Lunar and Planetary Science Conference* [[link]](<https://www.hou.usra.edu/meetings/lpsc2023/pdf/2960.pdf>)
 - Bowles, Neil and Ehlmann, Bethany and Klima, Rachel and Calcutt, Simon and Evans, Rory and **Klesh, A.** and Hoowe, Chris and Warren, Tristram and Shirley, Katherine and Curtis, Rowan and Bennett, Lee and Scire, Elena and Fogg, Robert and Furlan, Elise and Llamas, Jake and Seybold, Calina and Elkington, Nicholas and Blane, Diana and Dickson, Jay and Donaldson Hanna, Kerri and Edwards, Christopher and Green, Robert and House, Martha and Pieters, Carle and Thompson, David (2023). "The Lunar Thermal Mapper Instrument for the Lunar Trailblazer Mission." *European Lunar Symposium*
 - Bowles, Neil and Ehlmann, Bethany and Klima, Rachel and Calcutt, Simon and Evans, Rory and **Klesh, A.** and Hoowe, Chris and Warren, Tristram and Shirley, Katherine and Curtis, Rowan and Bennett, Lee and Scire, Elena and Fogg, Robert and Furlan, Elise and Llamas, Jake and Seybold, Calina and Elkington, Nicholas and Blane, Diana and Dickson, Jay and Donaldson Hanna, Kerri and Edwards, Christopher and Green, Robert and House, Martha and Pieters, Carle and Thompson, David (2023). "The Lunar Trailblazer Mission: Understanding the Moon's Water." *Fourth Workshop on Thermal Models for Planetary Science* [[link]](<https://www.cosmos.esa.int/documents/10435014/10435033/TherMoPSIV-abstract-book.pdf/6049c952-02aa-69a9-1bc7-3257561defef?t=1682514684648>)
 - {Shank}, Timothy M. and {Machado}, Casey and {Smith}, Russell and {Kapit}, Jason and {Michel}, Anna and {Nuzzio}, Donald B. and {Heyl}, Taylor P. and {Naklicki}, Victor and {Klesh}, Andrew (2022). "{Advancing Autonomous Underwater Vehicle Robotics for Full-Ocean Depth Exploration and Research}." *AGU Fall Meeting Abstracts* [[link]](<https://ui.adsabs.harvard.edu/abs/2022AGUFMOS14B..01S>)
 - Hardgrove, Craig J and Starr, Richard D and Prettyman, Thomas H and Lazbin, Igor and Johnson, Erik Bjorn and Roebuck, Bob and DuBois, Joe and Struebel, Nathaniel and Colaprete, Anthony and Hailey, Patrick and **Klesh, A.T.** and others (2022). "The Lunar Polar Hydrogen Mapper (LunaH-Map) Mission, Launch and Early Operations." *AGU Fall Meeting Abstracts* [[link]](<https://ui.adsabs.harvard.edu/abs/2022AGUFM>)
 - Ehlmann, Bethany L and Klima, Rachel L and Bennett, C Lee and Blaney, D and Bowles, Neil and Calcutt, S and Dickson, J and Donaldson Hanna, K and Edwards, Christopher S and Green, R and **Klesh, A.T.** and others (2022). "Lunar Trailblazer: A Pioneering Smallsat for Lunar Water and Lunar Geology." *American Geophysical Union* [[link]](<https://agu.confex.com/agu/fm22/meetingapp.cgi/Paper/1167201>)
 - Ehlmann, Bethany L and Klima, Rachel L and Bennett, C Lee and Blaney, D and Bowles, Neil and Calcutt, S and Dickson, J and Donaldson Hanna, K and Edwards, Christopher S and Green, R and **Klesh, A.T.** and others (2021). "Lunar Trailblazer: A Pioneering Smallsat for Lunar Water and Lunar Geology." *NASA Exploration Science Forum & European Lunar Symposium* [[link]](<https://www.youtube.com/watch?v=>)
 - Malphrus, Benjamin K and Freeman, Anthony and Staehle, Robert and **Klesh, A.T.** and Walker, Roger (2021). "Interplanetary CubeSat Missions." *Cubesat Handbook* [[link]](<https://www.sciencedirect.com/book/978012> handbook)
 - Ehlmann, Bethany L and Klima, Rachel L and Bennett, C Lee and Blaney, D and Bowles, Neil and Calcutt, S and Dickson, J and Donaldson Hanna, K and Edwards, Christopher S and Green, R and **Klesh, A.T.** and others (2021). "Lunar Trailblazer: A Pioneering Smallsat for Lunar Water and Lunar Geology." *Lunar and Planetary Science Conference* [[link]](<https://www.hou.usra.edu/meetings/lpsc2021/pdf/1740.pdf>)
 - Hallinan, Gregg and Burns, Jack and Lux, Jim and Romero-Wolf, Andres and Teitelbaum, Lawrence and

- Chang, Tzu-Ching and Kocz, Jonathon and Bowman, Judd and MacDowall, Robert and Kasper, Justin and **Klesh, A.T.** and others (2021). "FARSIDE: A Low Radio Frequency Interferometric Array on the Lunar Farside." *Bulletin of the American Astronomical Society* [\[\[link\]\]](https://baas.aas.org/pub/2021n4i379/release)(https://baas.aas.org/pub/2021n4i379/release)
- Krasner, Sanford M and Bruvold, Kristoffer and Call, Jared A and Fieseler, Paul D and **Klesh, A.T.** and Michael Kobayashi, M and Lay, Norman E and Lim, Ryan S and Morabito, David D and Oudrhiri, Kamal and others (2021). "Reconstruction of Entry, Descent, and Landing Communications for the InSight Mars Lander." *Journal of Spacecraft and Rockets* [\[\[link\]\]](https://arc.aiaa.org/doi/10.2514/1.A34892)(https://arc.aiaa.org/doi/10.2514/1.A34892)
 - Karrenbach, Martin and Shen, Zhichao and Li, Zefeng and Cole, Steve and Williams, Ethan and Klesh, Andy and Wang, Xin and Zhan, Zhongwen and LaFlame, Lisa and Yartsev, Victor and others (2020). "Rapid deployment of distributed acoustic sensing systems to track earthquake activity." *SEG Technical Program Expanded Abstracts 2020* [\[\[link\]\]](https://library.seg.org/doi/10.1190/segam2020-3426905.1)(https://library.seg.org/doi/10.1190/segam2020-3426905.1)
 - Ehlmann, Bethany L and Klima, Rachel L and Bennett, C Lee and Blaney, Diana L and Bowles, Neil E and Calcutt, Simon B and Dickson, James L and Hanna, Kerri Donaldson and Edwards, Christopher S and Evans, Rory and **Klesh, A.T.** and others (2020). "Lunar Trailblazer: A Pioneering Small Satellite for Lunar Water and Lunar Geology." *American Geophysical Union* [\[\[link\]\]](https://ui.adsabs.harvard.edu/abs/2020AGUFMP060...01H/abstract)(https://ui.adsabs.harvard.edu/abs/2020AGUFMP060...01H/abstract)
 - Aguzzi, Jacopo and Flexas, María del Mar and Flogel, Sascha and Lo Iacono, Claudio and Tangherlini, Michael and Costa, Corrado and Marini, Simone and Bahamon, Nixon and Martini, Severine and Fanelli, Emanuela and **Klesh, A.T.** and others (2020). "Exo-ocean exploration with Deep-Sea Sensor and Platform Technologies." *Astrobiology* [\[\[link\]\]](https://www.liebertpub.com/doi/10.1089/ast.2019.0001)(https://www.liebertpub.com/doi/10.1089/ast.2019.0001)
 - Okino, Clayton and Abraham, Douglas and Baker, John and Finley, Susan and Gao, Jay and Kahan, Daniel and **Klesh, A.T.** and Krajewski, Joel and Lay, Norman and Malhotra, Shan and others (2020). "Opportunistic Arraying." *IEEE Aerospace Conference* [\[\[link\]\]](https://ieeexplore.ieee.org/document/9172675)(https://ieeexplore.ieee.org/document/9172675)
 - Oudrhiri, Kamal and Yang, Oscar and Buccino, Dustin and Kahan, Daniel and Withers, Paul and Tortora, Paolo and Matousek, Steve and Lay, Norman and Lazio, Joseph and Krajewski, Joel and **Klesh, A.T.** and others (2020). "MarCO Radio Occultation: How the First Interplanetary Cubesat Can Help Improve Future Missions." *IEEE Aerospace Conference* [\[\[link\]\]](https://ieeexplore.ieee.org/document/9172734)(https://ieeexplore.ieee.org/document/9172734)
 - Ehlmann, Bethany L and Klima, Rachel L and Bennett, C Lee and Blaney, Diana L and Bowles, Neil E and Calcutt, Simon B and Dickson, James L and Hanna, Kerri Donaldson and Edwards, Christopher S and Evans, Rory and **Klesh, A.T.** and others (2020). "Lunar Trailblazer: A Pioneering Small Satellite for Lunar Water and Lunar Geology." *NASA Solar System Exploration Research Virtual Institute* [\[\[link\]\]](https://www.youtube.com/watch?v=KhIbNDzD2SM)(https://www.youtube.com/watch?v=KhIbNDzD2SM)
 - Hardgrove, Craig J and Starr, Richard D and Prettyman, Thomas H and Lazbin, Igor and Johnson, Erik Bjorn and Roebuck, Bob and DuBois, Joe and Struebel, Nathaniel and Colaprete, Anthony and Hailey, Patrick and **Klesh, A.T.** and others (2020). "The Lunar Polar Hydrogen Mapper (LunaH-Map) Mission: Low-Altitude Hydrogen Mapping of the Lunar South Pole With a Very Small Spacecraft." *AGU Fall Meeting Abstracts* [\[\[link\]\]](https://ui.adsabs.harvard.edu/abs/2020AGUFMP060...01H/abstract)(https://ui.adsabs.harvard.edu/abs/2020AGUFMP060...01H/abstract)
 - **Klesh, A.T.** and Krajewski, Joel and Baker, John (2019). "MarCO: Trailblazing Interplanetary Small Science." *4th COSPAR Symposium on Small Spacecraft* [\[\[link\]\]](https://program.eventact.com/Agenda/Lecture/2019090401)(https://program.eventact.com/Agenda/Lecture/2019090401)
 - Shank, TM and Machado, C and German, CR and Bowen, A and Leichty, JM and **Klesh, A.T.** and Smith, RG and Hand, KP (2019). "Development of a New Class of Autonomous Underwater Vehicle (AUV), Orpheus, for the Exploration of Ocean World Analogues." *Lunar Planetary Institute* [\[\[link\]\]](https://ui.adsabs.harvard.edu/abs/2019LPICo2168.6021S/abstract)(https://ui.adsabs.harvard.edu/abs/2019LPICo2168.6021S/abstract)
 - Sternberg, David and Essmiller, John and Colley, Cody and **Klesh, A.T.** and Krajewski, Joel (2019). "Attitude Control System for the Mars Cube One Spacecraft." *IEEE Aerospace Conference* [\[\[link\]\]](https://ieeexplore.ieee.org/document/9172675)(https://ieeexplore.ieee.org/document/9172675)
 - **Klesh, A.T.** and Baker, John and Krajewski, Joel (2019). "MarCO: Flight Review and Lessons Learned." *Small Sat* [\[\[link\]\]](https://digitalcommons.usu.edu/smallsat/2019/all2019/276/)(https://digitalcommons.usu.edu/smallsat/2019/all2019/276/)
 - Oudrhiri, Kamal and Yang, Yu-Ming and Withers, Paul and Buccino, Dustin and Lay, Norman and Tortora, Paolo and Kahan, Daniel S and Lazio, Joseph and **Klesh, A.T.** and Baker, John and others

- (2019). "Radio Occultation Results from MarCO, The First Interplanetary CubeSat Mission." *AGU Fall Meeting Abstracts* [[link]](<https://ui.adsabs.harvard.edu/abs/2019AGUFM.A43F..06O/abstract>)
- Robo, Ellen Trim and Labedz, Celeste Ritter and Williams, Ethan F and **Klesh, A.T.** and Panning, Mark P and Marandi, Alireza and Zhan, Zhongwen (2019). "DAS on Ice: Insights gained from deploying several Kilometers of Fiber Optic Cable on Taku Glacier, Alaska." *AGU Fall Meeting Abstracts* [[link]](<https://ui.adsabs.harvard.edu/abs/2019AGUFM.S23D0665R/abstract>)
- Fenty, Ian G and Woodward, Gail and Castano, Rebecca and **Klesh, A.T.** (2019). "Heterogenous System for In Situ Measurements from within Ice Shelf Cavities to Improve Predictions of Sea Level Rise." *AGU Fall Meeting Abstracts* [[link]](<https://ui.adsabs.harvard.edu/abs/2018AGUFM.C21B1314C/abstract>)
- Li, Zefeng and Zhan, Zhongwen and Williams, Ethan and **Klesh, A.T.** (2019). "Monitor Aftershocks of the 2019 Ridgecrest Earthquakes with DAS Arrays." *AGU Fall Meeting Abstracts*
- Baker, John and Colley, Cody N and Essmiller, John C and **Klesh, A.T.** and Krajewski, Joel A and Sternberg, David C (2019). "MarCO: The First Interplanetary CubeSats." *European Planetary Science Conference / Division of Planetary Science* [[link]](<https://meetingorganizer.copernicus.org/EPSC-DPS2019/EPSC-DPS2019-2009-2.pdf>)
- Krasner, Sandy and Bruvold, Kristoffer and **Klesh, A.T.** and Kobayashi, Masatoshi and Call, Jared and Lim, Ryan and Oudrhiri, Kamal and Lay, Norman and Wallace, Mark and Morabito, David and others (2019). "EDL Communications Featuring MarCO CubeSats." *International Planetary Probe Workshop*
- Klesh, A.T.** (2019). "MarCO: Flight Results from the First Interplanetary CubeSat Mission." *International Astronautical Congress* [[link]](<https://www.iafastro.org/events/iac/iac-2019/technical-programme/keynotes/25-october-2019-morning-session/b48-marco-flight-results-from-the-first-interplanetary-cubesat-mission.html>)
- Klesh, A.T.** and Clement, Brian and Colley, Cody and Essmiller, John and Forgette, Daniel and Krajewski, Joel and Marinan, Anne and Martin-Mur, Tomas (2018). "MarCO: Early Operations of the First CubeSats to Mars." *SmallSat Conference* [[link]](<https://digitalcommons.usu.edu/smallsat/2018/all2018/474/>)
- Klesh, A.T.** and Krajewski, Joel (2018). "Early Flight Results of the MarCO Spacecraft." *Low Cost Planetary Missions Workshop*
- Tardivel, Simon and **Klesh, A.T.** and Campagnola, Stefano (2018). "Technology Enabling Interplanetary Trajectories for NanoSpacecraft." *Journal of Spacecraft and Rockets* [[link]](<https://arc.aiaa.org/doi/10.2514/1.107000>)
- Klesh, A and Krajewski, J (2018). "MarCO: Mars Cube One—Lessons Learned from Readyng the First Interplanetary Cubesats for Flight." *49th Annual Lunar and Planetary Science Conference* [[link]](<https://www.hou.usra.edu/meetings/lpsc49/>)
- Clement, Brian and Benardini, James Nick and **Klesh, A.T.** and Newlin, Laura and Borenstein, Elan and Krajewski, Joel A and Li, Don (2018). "Planetary Protection Strategy and Implementation for MarCO, the First Interplanetary CubeSat Mission." *42nd COSPAR Scientific Assembly* [[link]](<https://ui.adsabs.harvard.edu/abs/2018cosp...42E.672C/abstract>)
- Klesh, A.T.** and Clement, Brian and Krajewski, Joel A (2018). "MarCO: Preparing for Interplanetary Flight with Small Spacecraft." *42nd COSPAR Scientific Assembly* [[link]](<https://ui.adsabs.harvard.edu/abs/2018cosp...42E.672C/abstract>)
- Castano, Rebecca and Boening, Carmen and Clark, Evan and Fenty, Ian G and Gierach, Michelle M and Hand, Kevin P and Khazendar, Ala and **Klesh, A.T.** and Leichty, John M and Limonadi, Daniel and others (2018). "Heterogenous System for In Situ Measurements from within Ice Shelf Cavities to Improve Predictions of Sea Level Rise." *AGU Fall Meeting Abstracts* [[link]](<https://ui.adsabs.harvard.edu/abs/2018AGUFM.C21B1314C/abstract>)
- Klesh, A.T.** and Krajewski, Joel (2018). "MarCO: Flight Status and Lessons Learned." *AGU Fall Meeting Abstracts* [[link]](<https://ui.adsabs.harvard.edu/abs/2018AGUFM.A44G..03K/abstract>)
- Boening, Carmen and Fenty, Ian G and Woodward, Gail and **Klesh, A.T.** and Castano, Rebecca and Limonadi, Daniel and Schodlok, Michael and Gierach, Michelle M and Hand, Kevin P (2018). "A New Two-Vehicle Autonomous System for Mapping Ice Shelf Cavities." *Ocean Sciences Meeting*
- Klesh, A.T.** (2018). "MarCO: Mars Cube One." *CubeSat Developer's Workshop* [[link]](<https://www.youtube.com/watch?v=Ug8v0u3t800>)
- Ehlmann, BL and Klesh, A and Alsedairy, T and Dekany, R and Dickson, J and Edwards, C and Forget, F and Fraeman, A and McCleese, D and Murchie, S and others (2018). "Mars Nano Or-

- biter: A CubeSat for Mars System Science.” *49th Annual Lunar and Planetary Science Conference* [[link]](<https://www.hou.usra.edu/meetings/lpsc2018/eposter/2818.pdf>)
- Schoolcraft, Josh and **Klesh, A.T.** and Werne, Thomas (2017). ”MarCO: Interplanetary Mission Development on a CubeSat Scale.” *Space Operations: Contributions from the Global Community* [[link]](<https://arc.aiaa.org/doi/10.2514/6.2017-2491>)
 - Lightholder, Jack and Thoesen, Andrew and Adamson, Eric and Jakubowski, Jeremy and Nallapu, Ravi and Smallwood, Sarah and Raura, Laksh and **Klesh, A.T.** and Asphaug, Erik and Thangavelautham, Jekan (2017). ”Asteroid Origins Satellite (AOSAT) I: An On-Orbit Centrifuge Science Laboratory.” *Acta Astronautica* [[link]](<https://www.sciencedirect.com/science/article/abs/pii/S0094576515301521>)
 - Asphaug, Erik and Thangavelautham, Jekan and **Klesh, A.T.** and Chandra, Aman and Nallapu, Ravi and Raura, Laksh and Herreras-Martinez, Mercedes and Schwartz, Stephen (2017). ”A CubeSat Centrifuge for Long Duration Milligravity Research.” *NPJ Microgravity* [[link]](<https://www.nature.com/articles/s41526-017-0021-0>)
 - Tardivel, Simon and **Klesh, A.T.** and Campagnola, Stefano (2017). ”Average Daily Radiation Dose as a Function of Altitude, Latitude, and Shielding.” *Journal of Spacecraft and Rockets* [[link]](<https://arc.aiaa.org/doi/10.2514/6.2017-1010>)
 - Ehlmann, Bethany and **Klesh, A.T.** and Alsedairy, Talal (2017). ”Mars NanoOrbiter: A CubeSat for Mars System Science.” *AAS/Division for Planetary Sciences Meeting* [[link]](<https://ui.adsabs.harvard.edu/abs/2017AAS...222.1010E>)
 - Hardgrove, C and Bell, J and Starr, R and Colaprete, T and Robinson, M and Drake, D and Lazbin, I and West, G and Johnson, E and Christian, J and **Klesh, A.T.** and others (2016). ”LunaH-Map (Lunar Polar Hydrogen Mapper): Orbital Neutron Spectroscopy from a 6U CubeSat.” *Committee on Space Research*
 - **Klesh, A.T.** (2016). ”MarCO: Ready for Flight.” *Proceedings of the 2016 CubeSat Developer’s Workshop*
 - Hardgrove, C and Bell, J and Starr, R and Colaprete, T and Robinson, M and Drake, D and Lazbin, I and West, G and Johnson, E and Christian, J and **Klesh, A.T.** and others (2016). ”The Lunar Polar Hydrogen Mapper (LunaH-Map) CubeSat Mission.” *International Astronautical Congress* [[link]](<https://iafastro.directory/iaac/paper/id/32804/summary/>)
 - Hardgrove, C and Bell, J and Starr, R and Colaprete, T and Robinson, M and Drake, D and Lazbin, I and West, G and Johnson, E and Christian, J and **Klesh, A.T.** and others (2016). ”The Lunar Polar Hydrogen Mapper (LunaH-Map) CubeSat Mission.” *47th Annual Lunar and Planetary Science Conference* [[link]](<https://www.hou.usra.edu/meetings/lpsc2016/pdf/2654.pdf>)
 - Baker, John and Castillo-Rogez, Julie and Bousquet, Pierre-W and Vane, Gregg and Komarek, Tomas and **Klesh, A.T.** (2016). ”Small Spacecraft for Planetary Science.” *41st COSPAR Scientific Assembly* [[link]](<https://www.science.gov/topicpages/e/earth+planetary+science>)
 - **Klesh, A.T.** (2016). ”INSPIRE and MarCO – Technology Development for the First Deep Space CubeSats.” *41st COSPAR Scientific Assembly* [[link]](<https://ui.adsabs.harvard.edu/abs/2016cosp...41E1014K/abstract>)
 - **Klesh, A.T.** and Krajewski, Joel (2016). ”MarCO–Ready for Launch.” *13th Annual CubeSat Developers Workshop, Cal Poly State Univ., San Luis Obispo, CA* [[link]](<http://mstl.atl.calpoly.edu/~workshop/archive/2018/S>)
 - Tardivel, Simon and **Klesh, A.T.** (2015). ”CubeSats for the Exploration and In-Situ Investigation of Asteroids.” *AAS Spaceflight Mechanics Conference*
 - **Klesh, A.T.** (2015). ”INSPIRE and Beyond: Deep Space CubeSats at JPL.” *Proceedings of the 2015 CubeSat Developer’s Workshop*
 - Santo, David and Rinker, George and **Klesh, A.T.** (2015). ”Adapting a Sophisticated Ground Data System for use by a Deep Space CubeSat Mission.” *Ground Systems Architectures Workshop* [[link]](<https://gsaw.org/wp-content/uploads/2015/03/2015s02santo.pdf>)
 - **Klesh, A.T.** (2015). ”Deep Space Small Satellites and INSPIRE.” *AstroRecon*
 - Tardivel, S and Canalias, E and Deleuze, M and **Klesh, A.T.** and Scheeres, DJ (2015). ”Landing MASCOT on Asteroid 1999 JU3: Solutions for Deploying Nanosats to Small-Body Surfaces.” *Lunar and*

Planetary Science Conference [[link]](<https://www.hou.usra.edu/meetings/lpsc2015/pdf/1182.pdf>)

- Castillo-Rogez, JC and Frick, A and **Klesh, A.T.** and Pavone, M and Nesnas, IA and Thomson, DR and Chien, SA and Boland, JS (2015). "Decadal Survey and Exploration Science with Next Generation Low Cost Small Platforms." *Spacecraft Reconnaissance of Asteroid and Comet Interiors* [[link]](<https://ui.adsabs.harvard.edu/abs/2015LPICo1829.6042C/abstract>)
- **Klesh, A.T.** and Krajewski, Joel (2015). "Marco: Cubesats to Mars in 2016." *SmallSat Conference* [[link]](<https://digitalcommons.usu.edu/smallsat/2015/all2015/16/>)
- Spangelo, Sara and Castillo-Rogez, Julie and Frick, Andy and Klesh, Andy and Sherwood, Brent (2015). "JPL's Advanced CubeSat Concepts for Deep Space Exploration." *SmallSat Conference* [[link]](<https://digitalcommons.usu.edu/smallsat/2015/all2015/16/>)
- Sherwood, Brent and Spangelo, Sara and Frick, Andreas and Castillo-Rogez, Julie and **Klesh, A.T.** and Wyatt, E Jay and Reh, Kim and Baker, John (2015). "Planetary CubeSats Come of Age." *International Astronautical Congress* [[link]](<https://iafastro.directory/iac/archive/browse/IAC-15/A3/5/30103/>)
- Spangelo, Sara and Castillo-Rogez, Julie and Frick, Andy and Klesh, Andy and Sherwood, Brent (2015). "JPL's Advanced CubeSat Concepts for Interplanetary Science and Exploration Missions." *CubeSat Developer's Workshop* [[link]](<http://mstl.atl.calpoly.edu/~workshop/archive/2015/Summer/Day%202/1130-Spangelo-JPL.pdf>)
- Matheus-Carnevali, P and Rohrsen, M and Williams, M and Love, G and Michaud, A and Adams, H and Priscu, J and Berisford, D and **Klesh, A.** and Leichty, John and Hand, Kevin and Murray, Allison (2014). "Methane Sources in Arctic Thermokarst Lake Sediments on the North Slope of Alaska." *Goldschmidt* [[link]](<https://goldschmidtabstracts.info/2014/1621.pdf>)
- Castillo-Rogez, Julie and Boland, JS and Chien, SA and Frick, Andy and Kahn, Cindy and Kahn, Peter and **Klesh, A.T.** and Nesnas, Issa and Pavone, Marco and Staehle, Robert and Thompson, David (2014). "Decadal and SKG Science with Next Generation Low Cost Small Platforms." *Proceedings of the 2014 SSERVI Exploration Science Forum*
- **Klesh, A.T.** and Baker, John and Bellardo, John and Castillo-Rogez, Julie and Cutler, James and Halatek, Lauren and Lightsey, E Glenn and Murphy, Neil and Raymond, Carol (2014). "INSPIRE: Interplanetary NanoSpacecraft Pathfinder in a Relevant Environment." *Proceedings of the 2014 International Astronautical Congress* [[link]](<https://iafastro.directory/iac/paper/id/26818/abstract-pdf/IAC-14,B4,8,1,x26818.brief.pdf?2014-04-10.17:12:27>)
- **Klesh, A.T.** and Baker, John and Bellardo, John and Castillo-Rogez, Julie and Cutler, James and Halatek, Lauren and Lightsey, E Glenn and Murphy, Neil and Raymond, Carol (2014). "INSPIRE: Interplanetary NanoSpacecraft Pathfinder in a Relevant Environment." *Proceedings of the 2014 CubeSat Developer's Workshop*
- Liewer, Paulett C and **Klesh, A.T.** and Lo, Martin W and Murphy, Neil and Staehle, Robert L and Angelopoulos, Vassilis and Anderson, Brian D and Arya, Manan and Pellegrino, Sergio and Cutler, James W and others (2014). "A Fractionated Space Weather Base at L5 using CubeSats and Solar Sails." *Advances in Solar Sailing* [[link]](https://link.springer.com/chapter/10.1007/978-3-642-34907-2_19)
- Quach, William L and DeForrest, Lloyd and **Klesh, A.T.** and Schoolcraft, Josh (2014). "Adapting a Large-Scale Multi-Mission Ground System for Low-Cost CubeSats." *SpaceOps 2014 Conference* [[link]](<https://arc.aiaa.org/doi/10.2514/6.2014-1634>)
- Staehle, Robert L and Anderson, Brian and Betts, Bruce and Blaney, Diana and Chow, Channing and Friedman, Louis and Hemmati, Hamid and Jones, Dayton and **Klesh, A.T.** and Liewer, Paulett and others (2013). "Interplanetary CubeSats: Opening the Solar System to a Broad Community at Lower Cost." *Journal of Small Satellites* [[link]](<https://jossonline.com/storage/2014/12/0201-Interplanetary-CubeSats-Opening-the-Solar-System-to-a-Broad-Community-at-Lower-Cost.pdf>)
- Liewer, Paulett and **Klesh, A.T.** and Lo, Martin W and Murphy, Neil and Staehle, Robert and Vourlidas, Angelo and Cutler, James and Lightsey, E Glenn (2013). "A Fractionated Space Weather Base at L5 Using CubeSats and Solar Sails." *Solar Physics Division Meeting*

- Castillo-Rogez, Julie and **Klesh, A.T.** and Kahn, Peter and Staehle, Robert and Nesnas, Issa and Pavone, Marco (2013). "Next Generation SmallSat – Dare to Explore Where No Craft Has Gone Before." *Proceedings of the 2013 Interplanetary Small Satellite Conference*
- Halatek, Lauren and Castillo-Rogez, Julie and **Klesh, A.T.** (2013). "CubeSats and Europa: Focused Science with Disposable Spacecraft." *Proceedings of the 2013 Low Cost Planetary Missions Conference*
- Castillo-Rogez, Julie and **Klesh, A.T.** and Kahn, Peter and Staehle, Robert and Nesnas, Issa and Pavone, Marco (2013). "Next Generation SmallSat – Dare to Explore Where No Craft Has Gone Before." *Proceedings of the 2013 Low Cost Planetary Missions Conference*
- **Klesh, A.T.** and Baker, John and Bellardo, John and Castillo-Rogez, Julie and Cutler, James and Halatek, Lauren and Lightsey, E Glenn and Murphy, Neil and Raymond, Carol (2013). "INSPIRE: Interplanetary NanoSpacecraft Pathfinder in a Relevant Environment." *Proceedings of the 2013 Low Cost Planetary Missions Conference*
- **Klesh, A.T.** and Baker, John and Bellardo, John and Castillo-Rogez, Julie and Cutler, James and Halatek, Lauren and Lightsey, E Glenn and Murphy, Neil and Raymond, Carol (2013). "INSPIRE: Interplanetary NanoSpacecraft Pathfinder in a Relevant Environment." *Proceedings of the 2013 CubeSat Developer's Workshop*
- Cutler, James and **Klesh, A.T.** (2013). "A Precursor Mission for Deep Space Cubesats." *2nd International Workshop on LunarCubes*
- **Klesh, A.T.** and Gustafson, Eric and Strange, Nathan and Wilcox, Bryan (2013). "Simulation and Analysis of a Phobos-Anchored Tether." *AAS Spaceflight Mechanics Meeting*
- Spangelo, Sara C and Bennett, Matthew W and Meinzer, Daniel C and **Klesh, A.T.** and Arlas, Jessica A and Cutler, James W (2013). "Design and Implementation of the GPS Subsystem for the Radio Aurora eXplorer." *Acta Astronautica* [[link]](<https://www.sciencedirect.com/science/article/abs/pii/S0094576512004894>)
- **Klesh, A.T.** and Castillo-Rogez, Julie C (2013). "Applicability of Nano-Spacecraft to Deep Space Explorationeep Space Exploration." *44th Annual Lunar and Planetary Science Conference*
- **Klesh, A.T.** and Baker, John D and Bellardo, John and Castillo-Rogez, Julie and Cutler, James and Halatek, Lauren and Lightsey, E Glenn and Murphy, Neil and Raymond, Carol (2013). "INSPIRE: Interplanetary NanoSpacecraft Pathfinder in Relevant Environment." *AIAA SPACE 2013 Conference and Exposition* [[link]](<https://arc.aiaa.org/doi/10.2514/6.2013-5323>)
- Marchis, F and Castillo-Rogez, J and Jenkins, J and Dissly, R and Klesh, A (2013). "Origami NanoSat Telescopes: Planetary Astronomy's Future unfolds." *European Planetary Science Congress*
- Berisford, Daniel F and Leichty, John and **Klesh, A.** and Hand, Kevin P (2013). "Remote Under-Ice Roving in Alaska with the Buoyant Rover for Under-Ice Exploration." *AGU Fall Meeting Abstracts* [[link]](<https://ui.adsabs.harvard.edu/abs/2013AGUFM.C13C0684B/abstract>)
- Leichty, John M and **Klesh, A.T.** and Berisford, Daniel F and Matthews, Jaret B and Hand, Kevin P (2013). "Positive-Buoyancy Rover for Under Ice Mobility." *Magazine* [[link]](https://www.techbriefs.com/component_npo-48863)
- Lazio, Joseph and Staehle, Robert L and **Klesh, A.T.** and Romero-Wolf, Andrew and Blaney, Diana and Hemmati, Hamid and Jones, Dayton and Liewer, Paulett and Lo, Martin Wen-Yu and Mouroulis, Pantazis and others (2013). "Interplanetary CubeSat Architecture and Missions." *LunarCubes*
- Liewer, Paulett C and **Klesh, A.T.** and Lo, Martin W and Murphy, Neil and Staehle, Robert L and Angelopoulos, Vassilis and Anderson, Brian D and Arya, Manan and Pellegrino, Sergio and Cutler, James W and others (2013). "A Fractionated Space Weather Base at L5 using CubeSats \{\}& Solar Sails." *Proc. Interplanet. Small Satell. Conf., Pasadena, CA*
- Staehle, Robert and Puig-Suari, Jordi and Svitek, Tomas and Friedman, Louis and Blaney, Diana and **Klesh, A.T.** (2012). "Interplanetary CubeSats: Some Missions Feasible Sooner Than Expected." *1st Interplanetary CubeSat Workshop*
- Staehle, Robert and Blaney, Diana and Hemmati, Hamid and Jones, Dayton and **Klesh, A.T.** and

- Lazio, Joesph and Liewer, Paulett and Lo, Martin Wen-Yu and Mouroulis, Pantazis and Murphy, Neil and Pingree, Paula and others (2012). "Missions Enabled by an Interplanetary CubeSat Architecture." *Proceedings of the 2012 CubeSat Developer's Workshop*
- **Klesh, A.T.** and Castillo-Rogez, Julie (2012). "Applicability of NanoSatellites to Small Body Exploration." *Proceedings of the 2012 CubeSat Developer's Workshop*
 - Springmann, John C and Sloboda, Alexander J and **Klesh, A.T.** and Bennett, Matthew W and Cutler, James W (2012). "The Attitude Determination System of the RAX Satellite." *Acta Astronautica* [[link]](<https://www.sciencedirect.com/science/article/abs/pii/S0094576512000252>)
 - Spangelo, Sara C and Cutler, James W and **Klesh, A.T.** and Boone, DR (2012). "Models and Tools to Evaluate Space Communication Network Capacity." *IEEE Transactions on Aerospace and Electronic Systems* [[link]](<https://ieeexplore.ieee.org/document/6237598>)
 - **Klesh, A.T.** and Castillo-Rogez, JC (2012). "Secondary NanoSpacecraft Survey of the Martian Moons." *Concepts and Approaches for Mars Exploration* [[link]](<https://www.lpi.usra.edu/meetings/marsconcepts2012/pdf/4>)
 - **Klesh, A.T.** and Castillo-Rogez, JC (2012). "Applications of NanoSatellites to Small-Body Exploration." *Asteroids, Comets and Meteors* [[link]](<https://www.lpi.usra.edu/meetings/acm2012/pdf/6478.pdf>)
 - Staehle, RL and Betts, B and Friedman, LD and Hemmati, H and **Klesh, A.T.** and Lo, MW and Mouroulis, P and Pingree, PJ and Puig-Suari, J and Svitek, T and others (2012). "Solar Sails: From Interplanetary CubeSats for Remote Sensing to Phobos Sample Return to Clipper Ships for Pre-Placing Human Exploration Equipment in Mars Orbit." *Concepts and Approaches for Mars Exploration* [[link]](<https://www.lpi.usra.edu/meetings/marsconcepts2012/pdf/4123.pdf>)
 - Strange, NJ and **Klesh, A.T.** and Marrese-Reading, CM and Oh, DY and Ziemer, JK and McElrath, TP and Landau, DF and Grebow, DJ (2012). "Interplanetary Sample Canister for Mars Sample Return." *Concepts and Approaches for Mars Exploration* [[link]](<https://www.lpi.usra.edu/meetings/marsconcepts2012/pdf/4>)
 - **Klesh, A.T.** and Castillo-Rogez, JC (2012). "Applicability of Nanosatellites to Deep Space Exploration." *Lunar and Planetary Science Conference* [[link]](<https://www.lpi.usra.edu/meetings/lpsc2012/pdf/2326.pdf>)
 - **Klesh, A.T.** and Cutler, James W and Atkins, Ella M (2012). "Cyber-Physical Challenges for Space Systems." *2012 IEEE/ACM Third International Conference on Cyber-Physical Systems* [[link]](<https://ieeexplore.ieee.org/document/6237598>)
 - **Klesh, A.T.** and Castillo-Rogez, Julie (2012). "Applications of NanoSats to Small Body Exploration." *AIAA SPACE 2012 Conference & Exposition* [[link]](<https://arc.aiaa.org/doi/10.2514/6.2012-5326>)
 - Staehle, Robert and Blaney, Diana and Hemmati, Hamid and Jones, Dayton and **Klesh, A.T.** and Liewer, Paulett and Lazio, Joseph and Wen-Yu Lo, Martin and Mouroulis, Pantazis and Murphy, Neil and others (2012). "Interplanetary CubeSat Architecture and Missions." *AIAA SPACE 2012 Conference & Exposition* [[link]](<https://arc.aiaa.org/doi/10.2514/6.2012-5218>)
 - **Klesh, A.T.** and Castillo, JC (2012). "Science-Driven NanoSats Design for Deep Space." *AGU Fall Meeting Abstracts* [[link]](<https://ui.adsabs.harvard.edu/abs/2012AGUFM.P31A1882K/abstract>)
 - Bekker, DL and Bryk, M and DeLucca, J and Franklin, B and Hancock, B and **Klesh, A.T.** and Meehan, C and Meshkaty, N and Nichols, J and Pingree, P and others (2012). "GRIFEX Payload Data System Architecture for On-Orbit Focal Plane Array Evaluation." *AGU Fall Meeting Abstracts* [[link]](<https://ui.adsabs.harvard.edu/abs/2012AGUFMIN31C1514B/abstract>)
 - Pingree, P and Bekker, DL and Bryk, M and DeLucca, J and Franklin, B and Hancock, B and **Klesh, A.T.** and Meehan, C and Meshkaty, N and Nichols, J and others (2012). "COVE, MARINA, and the Future of On-Board Processing (OBP) Platforms for CubeSat Science Missions." *AGU Fall Meeting Abstracts* [[link]](<https://ui.adsabs.harvard.edu/abs/2012AGUFMIN31C1512P/abstract>)
 - Berisford, DF and Leichty, JM and **Klesh, A.T.** and Matthews, JB and Hand, KP (2012). "Buoyant Rover for Under-Ice Exploration." *AGU Fall Meeting Abstracts* [[link]](<https://ui.adsabs.harvard.edu/abs/2012AGUFM.P31A1882K/abstract>)
 - **Klesh, A.T.** and Castillo-Rogez, Julie C (2012). "Nano-Satellite Secondary Spacecraft on Deep Space Missions." *GLEX* [[link]](<https://dataverse.jpl.nasa.gov/api/access/datafile/52179?gbrecs=true>)
 - **Klesh, A.T.** and Angelopoulos, Vassilis and Betts, Bruce and Bidy, Chris and Caron, Ryan and Cutler,

- James and Desai, Mihir and Gustafson, Eric and Friedman, Louis and Hamlin, Louise and others (2012). "SolWise: Sailing On Light With Interplanetary Science and Exploration." *Annual AIAA/USU Conference on Small Satellites* [[link]](<https://dataverse.jpl.nasa.gov/dataset.xhtml?persistentId=hdl:2014/43168>)
- Staehle, Robert L and Anderson, Brian and Betts, Bruce and Blaney, Diana and Chow, Channing and Friedman, Louis and Hemmati, Hamid and Jones, Dayton and **Klesh, A.T.** and Liewer, Paulett and others (2012). "Interplanetary CubeSats: Opening the Solar System to a Broad Community at Lower Cost." *Conference*
- Cocaud, Cedric and **Klesh, A.T.** and Kubota, Takashi (2011). "Reducing the Uncertainty of Hayabusa's Landing Position on Itokawa." *Proceedings of the 2011 International Astronautical Congress* [[link]](<https://iafastro.directory/iaac/2011/C1/7/9959/>)
- Klesh, A.T.** (2011). "Improving Surface Exploration for Small-Body Objects." *Proceedings of the 2011 International Primitive Body Exploration Working Group Workshop*
- Klesh, A.T.** and Yoshimitsu, Tetsuo and Kubota, Takashi (2011). "Improved Navigation Techniques for Asteroid Landers and Impactors." *Proceedings of the 2011 IAA Planetary Defense Conference*
- Klesh, A.T.** and Mori, Osamu and Sawada, Hirotaka and Tsuda, Yuichi and Matunaga, Saburo and Kimura, Shinichi (2011). "Trajectory Reconstruction of the IKAROS DCAM2." *21st AAS/AIAA Space Flight Mechanics Meeting* [[link]](<https://tus.elsevierpure.com/en/publications/trajec-tory-reconstruction-of-the-ikaros-dcam2>)
- Klesh, A.T.** and Yoshimitsu, Tetsuo and Kubota, Takashi and others (2011). "Autonomous Path Planning for Multi-Vehicle Wide-Baseline Stereo." *The 20th Workshop on JAXA Astrodynamics and Flight Mechanics*
- Klesh, A.T.** and Bellerose, Julie (2010). "Guidance and Control of Hoppers for Small Body Surface Ex-ploration." *Proceedings of the 2010 International Astronautical Congress* [[link]](<https://iafastro.directory/iaac/archive/2010/A3/5/6575/>)
- Klesh, A.T.** (2010). "Path Planning for Distributed Rover Science." *JAXA Space Exploration Center*
- Cutler, James and Bennett, Matthew and **Klesh, A.T.** and Bahcivan, Hasan and Doe, Rick (2010). "The Radio Aurora Explorer—A Bistatic Radar Mission to Measure Space Weather Phenomenon." *Ameri-can Institute of Aeronautics and Astronautics* [[link]](<https://digitalcommons.usu.edu/smallsat/2010/all2010/12/>)
- Spangelo, Sara and **Klesh, A.T.** and Cutler, James (2010). "Position and Time System for the RAX Small Satellite Mission." *AIAA/AAS Astrodynamics Specialist Conference* [[link]]([https://exploration.engin.umich.edu/content/uploads/sites/578/2022/08/Spangelo.et.al_2010b.pdf](https://exploration.engin.umich.edu/content/uploads/sites/578/2022/08/Spangelo_et.al_2010b.pdf))
- Senatore, Patrick and **Klesh, A.T.** and Zurbuchen, Thomas H and McKague, Darren and Cutler, James (2010). "Concept, Design, and Prototyping of XSAS: A High Power Extendable Solar Array for CubeSat Applications." *Proceedings of the 40th Aerospace Mechanisms Symposium* [[link]](<https://ntrs.nasa.gov/api/citations/2010-02-18000/main.pdf>)
- Klesh, A.T.** and Cutler, James (2010). "Exploiting the Link: Improving Satellite Communication Through Higher Elevation Links." *AIAA/AAS Astrodynamics Specialist Conference* [[link]](<https://arc.aiaa.org/doi/10.2514/6.2010-8269>)
- Trabert, Rachel and **Klesh, A.T.** and Senatore, Patrick and Martinchek, Patrick and Becker, Daniel and Chou, Andrew and Hoffman, Clark and Hoffman, Clark and McKay, Nathan and Nash, Jeremy and others (2010). "The Extendable Solar Array System: A Modular NanoSatellite Power System." *AIAA/AAS Astrodynamics Specialist Conference* [[link]](<https://arc.aiaa.org/doi/10.2514/6.2010-7654>)
- Jaffer, Ghulam and Nader, Ronnie and Koudelka, Otto and **Klesh, A.T.** and others (2010). "Using a Virtual Ground Station as a Tool for Supporting Higher Education." *61st International Astronautical Congress* [[link]](<https://iafastro.directory/iaac/paper/id/6982/summary/>)
- Jaffer, Ghulam and Nader, Ronnie and **Klesh, A.T.** and Koudelka, Otto (2010). "Project Agora: Simultaneously Downloading a Satellite Signal Around the World." *International Astronautical Federation* [[link]](<https://iafastro.directory/iaac/archive/browse/IAC-10/B2/5/6981/>)
- Klesh, A. T** and Kabamba, Pierre T and Girard, Anouck R (2009). "Optimal cooperative thermalling of

- unmanned aerial vehicles." *Optimization and Cooperative Control Strategies: Proceedings of the 8th International Conference on Cooperative Control and Optimization* [\[\[link\]\]](https://link.springer.com/chapter/10.1007/978-3-540-88063-9_20)(https://link.springer.com/chapter/10.1007/978-3-540-88063-9_20)
- **Klesh, A.T.** and Cutler, James (2009). "Exploiting the Link: Improving CubeSat Communications through the use of Software Defined Radios." *Software Defined Radio Forum* [\[\[link\]\]](https://exploration.engin.umich.edu/content/uploads/sites/578/2022/08/AAS_2010.pdf)(https://exploration.engin.umich.edu/content/uploads/sites/578/2022/08/AAS_2010.pdf)
 - **Klesh, A.T.** and Jacobs, Shawn (2009). "A Comparison of Mars Precursor Missions: Lunar and Arctic." *International Astronautical Congress* [\[\[link\]\]](https://iafastro.directory/iac/paper/id/3112/summary/)(<https://iafastro.directory/iac/paper/id/3112/summary/>)
 - Cutler, James and Gilchrist, Brian and McKague, Darren and Bennett, Matthew and Dontchev, Kiko and Heywood, Michael and **Klesh, A.T.** (2009). "Michigan Nanosat Pipeline." *CubeSat Conference*
 - Dontchev, Kiko and **Klesh, A.T.** and Bennett, Matthew and Cutler, James (2009). "In-Lab Testing for Attitude Determination and Control." *Proceedings of the 2009 CubeSat Developer's Workshop*
 - **Klesh, A.T.** and Bellerose, Julie (2009). "Asteroid Deflection through Rotational Fragmentation." *Proceedings of the 2009 IAA Planetary Defense Conference*
 - **Klesh, A.T.** (2009). "SHOTPUT Sample Return: Examining the Compositional Gradient of Small-Body Objects." *40th Annual Lunar and Planetary Science Conference* [\[\[link\]\]](https://www.lpi.usra.edu/meetings/lpsc2009/)(<https://www.lpi.usra.edu/meetings/lpsc2009/>)
 - Britt, Daniel and Abell, Paul and Ammannito, Eleonora and Asphaug, Erik and Aung, MiMi and Bell, Jim and Bellerose, Julie and Benna, Mehdi and Benner, Lance and Blewett, David and **Klesh, A.T.** and others (2009). "Asteroids." *Community White Paper to the Planetary Science Decadal Survey, 2011-2020* [\[\[link\]\]](https://solarsystem.nasa.gov/studies/172/small-bodies-community-white-paper-asteroids/)(<https://solarsystem.nasa.gov/studies/172/small-bodies-community-white-paper-asteroids/>)
 - **Klesh, A.T.** and Kabamba, Pierre T (2009). "Solar-Powered Aircraft: Energy-Optimal Path Planning and Perpetual Endurance." *AIAA Journal of Guidance, Control, and Dynamics* [\[\[link\]\]](https://arc.aiaa.org/doi/10.2514/6.2009-1893)(<https://arc.aiaa.org/doi/10.2514/6.2009-1893>)
 - Atkins, Ella and Eubank, Ryan and **Klesh, A.T.** (2009). "A Reconfigurable Flight Management System for Small-Scale Unmanned Air Systems." *AIAA Infotech@ Aerospace Conference and AIAA Unmanned... Unlimited Conference* [\[\[link\]\]](https://arc.aiaa.org/doi/abs/10.2514/6.2009-1893)(<https://arc.aiaa.org/doi/abs/10.2514/6.2009-1893>)
 - Hyun, Baro and Jackson, Justin and **Klesh, A.T.** and Girard, Anouck and Kabamba, Pierre (2009). "Robotic Exploration with Non-Isotropic Sensors." *AIAA Guidance, Navigation, and Control Conference* [\[\[link\]\]](https://arc.aiaa.org/doi/10.2514/6.2009-6267)(<https://arc.aiaa.org/doi/10.2514/6.2009-6267>)
 - **Klesh, A.T.** and Seagraves, Sheryl and Bennett, Matt and Boone, Dylan and Cutler, James and Bahcivan, Hasan (2009). "Dynamically Driven Helmholtz Cage for Experimental Magnetic Attitude Determination." *Advances in the Astronautical Sciences*
 - **Klesh, A.T.** and Kabamba, Pierre T and Girard, Anouck R (2009). "Optimal cooperative thermalling of unmanned aerial vehicles." *Optimization and Cooperative Control Strategies* [\[\[link\]\]](https://link.springer.com/chapter/10.1007/978-3-540-88063-9_20)(https://link.springer.com/chapter/10.1007/978-3-540-88063-9_20)
 - **Klesh, A.T.** and Kabamba, Pierre T and Girard, Anouck R (2009). "Optimal Path Planning for Uncertain Exploration." *2009 American Control Conference* [\[\[link\]\]](https://ieeexplore.ieee.org/abstract/document/516028)(<https://ieeexplore.ieee.org/abstract/document/516028>)
 - **Klesh, A.T.** and Taniguchi, Shintaro and Gadre, Rajeev and Wang, Jiuguang and Moskovitz, Nicholas and Smith, Ramsey and Kongpop, U and Burke, Caley and Springmann, Alessondra and Horodyskyj, Lev and others (2009). "SHOTPUT: A JPL Planetary Summer Science School study." *2009 IEEE Aerospace conference* [\[\[link\]\]](https://ieeexplore.ieee.org/abstract/document/4839323/footnotes?signout=success#footnotes)(<https://ieeexplore.ieee.org/abstract/document/4839323/footnotes?signout=success#footnotes>)
 - **Klesh, A.T.** (2009). "Optimal Exploration Systems." *Thesis* [\[\[link\]\]](https://deepblue.lib.umich.edu/handle/2027.41/100000)(<https://deepblue.lib.umich.edu/handle/2027.41/100000>)
 - Spangelo, Sara and Gilbert, Elmer and **Klesh, A.T.** and Kabamba, Pierre and Girard, Anouck (2009). "Periodic Energy-Optimal Path Planning for Solar-Powered Aircraft." *AIAA Guidance, Navigation, and Control Conference* [\[\[link\]\]](https://arc.aiaa.org/doi/10.2514/6.2009-6016)(<https://arc.aiaa.org/doi/10.2514/6.2009-6016>)
 - Hasan, Zahid and Faied, Mariam and **Klesh, A.T.** and Girard, Anouck (2009). "Distributed Heterogeneous Control of Mini-Flying Machines and Ground Robots." *AIAA Guidance, Navigation, and Control Conference*
 - Bellerose, Julie and **Klesh, A.T.** (2009). "Internal Asteroid Composition through Distributed Sensing."

Proceedings of the 2009 International Astronautical Congress [\[\[link\]\]\(https://iafastro.directory/iaac/paper/id/3833/su\)](https://iafastro.directory/iaac/paper/id/3833/su)

- Budney, C and Burke, C and Cartwright, M and Gadre, R and Horodyskyj, L and Klesh, A and Milam, K and Moskovitz, N and Oiler, J and Ostrowski, D and others (2008). "SHOTPUT: A Mission Proposal to Study Composition and Origins of Small Bodies in the Outer Solar System Through Fly-By and Impactor Science." *AGU Fall Meeting Abstracts* [\[\[link\]\]\(https://ui.adsabs.harvard.edu/abs/2008AGUFM.P51C1426B/abstract\)](https://ui.adsabs.harvard.edu/abs/2008AGUFM.P51C1426B/abstract)
- Springmann, Alessondra and Burke, C and Cartwright, M and Gadre, R and Horodyskyj, L and Klesh, A and Milam, K and Moskovitz, N and Oiler, J and Ostrowski, D and others (2008). "Designing a New Frontiers-class Trojan/Centaur Reconnaissance Mission: A JPL Planetary Science Summer School Study." *AAS/Division for Planetary Sciences Meeting Abstracts* \{\} # 40 [\[\[link\]\]\(https://ui.adsabs.harvard.edu/abs/2008AAS...40..400S\)](https://ui.adsabs.harvard.edu/abs/2008AAS...40..400S)
- **Klesh, A.T.** and Kabamba, Pierre T and Girard, Anouck R (2008). "Path Planning for Cooperative Time-Optimal Information Collection." *2008 American Control Conference* [\[\[link\]\]\(https://ieeexplore.ieee.org/document/4551444\)](https://ieeexplore.ieee.org/document/4551444)
- **Klesh, A.T.** and Girard, Anouck and Kabamba, Pierre (2008). "Real-time path Planning for Time-Optimal Exploration." *AIAA Guidance, Navigation and Control Conference and Exhibit* [\[\[link\]\]\(https://arc.aiaa.org/article/doi/10.2514/6.2007-6652\)](https://arc.aiaa.org/article/doi/10.2514/6.2007-6652)
- **Klesh, A.T.** and Macy, Daniel and Smith, A and Senatore, Patrick and Rooney, N and Wiebenga, Jonathan (2008). "SolarBubbles: An Autonomous Solar-Powered UAV." *AUVSI Conference*
- Hedrick, J Karl and Basso, Brandon and Love, Joshua and Girard, Anouck R and **Klesh, A.T.** (2008). "Control of Mobile Sensor Networks: A State-of-the-Art Review." *Dynamic Systems and Control Conference* [\[\[link\]\]\(https://asmedigitalcollection.asme.org/DSCC/proceedings-abstract/DSCC2008/43352/1491/334031\)](https://asmedigitalcollection.asme.org/DSCC/proceedings-abstract/DSCC2008/43352/1491/334031)
- **Klesh, A.T.** and Kabamba, Pierre (2007). "Energy-Optimal Path Planning for Solar-Powered Aircraft in Level Flight." *AIAA Guidance, Navigation and Control Conference and Exhibit* [\[\[link\]\]\(https://arc.aiaa.org/article/doi/10.2514/6.2007-6655\)](https://arc.aiaa.org/article/doi/10.2514/6.2007-6655)
- **Klesh, A.T.** and Kabamba, Pierre (2007). "Solar-Powered Unmanned Aerial Vehicles on Mars: Perpetual Endurance." *Proceedings of the 58th International Astronautical Congress, no. IAC-07-D3* [\[\[link\]\]\(https://www.sciencedirect.com/science/article/pii/S0022316707000013\)](https://www.sciencedirect.com/science/article/pii/S0022316707000013)
- **Klesh, A.T.** (2001). "Project Aria – Aria 1 – Observations of Passive Experiments in Space." *Missouri Space Grant Consortium*

For a complete list with abstracts and additional details, please visit: [View Full Publications Page &rsarr;](#)